IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-009

1. Topic (12 words max)

AR-Based Intelligent STEM Learning System for Students

2. Research group the project belongs to

MR - Mixed Reality

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

The conventional method of teaching the subjects of STEM in the primary schools uses textbooks and the teacher-centred approach. This has been shown to be less effective for the young minds. Most of the concepts, for instance, human body, planetary systems, or natural phenomena, are abstract, meaning that they cannot be described with static pictures (Ibáñez & Delgado-Kloos, 2018).

The main users who experience this problem include students aged 5-11 years from primary schools, as well as teachers and parents. The students have difficulties learning good foundational concepts in STEM, and the teachers have no interactive technology resources that can be useful in demonstrations within the classroom. According to Radianti et al. (2020), interactive learning environments have been found to boost learning comprehension and interest by students in STEM (Radianti et al., 2020).

Present learning platforms offer basic multimedia content support, which does not incorporate contextual interaction, intelligent feedback, and personalization. Many learning apps involving AR emphasize 3D views alone, failing to incorporate artificial intelligence that can review student interaction patterns to render personalized support (Akçayır & Akçayır, 2017).inclusive and effective education, along with inspiring innovations in education technology.

This problem has major social and economic implications. Weak STEM foundations reduce students' future career opportunities and negatively affect the development of a skilled workforce. From a technical perspective, many schools have not adopted advanced learning technologies due to cost and infrastructure limitations. Ethically, unequal access to interactive learning tools increases educational disparities among students.

Therefore, there is a strong need for an intelligent AR-based STEM learning system that integrates immersive visualization, voice-based assistance, AI-driven assessments, and gamification. Sustainable Development Goal 4 (Quality Education) by promoting inclusive, engaging, and effective learning experiences while fostering innovation in educational technology.

References

- Ibáñez, M. B., & Delgado-Kloos, C. (2018). Augmented reality for STEM learning: A systematic review. *Computers & Education*, 123, 109–123.
- Radianti, J., Majchrzak, T. A., Fromm, J., & Wohlgenannt, I. (2020). A systematic review of immersive virtual reality applications for higher education. *Education and Information Technologies*, 25(6), 4957–4980.
- Akçayır, M., & Akçayır, G. (2017). Advantages and challenges associated with augmented reality for education. *Educational Research Review*, 20, 1–11.

6. Brief description of Existing Research and Systems (300 words max)

Several AR-based learning applications such as **Google Expeditions AR, Merge Cube, and AR Flashcards** currently exist in the education domain. These systems primarily focus on providing basic 3D visualization to support learning. While they enhance student engagement to some extent, they lack intelligent interaction mechanisms and personalized learning features, which limits their overall educational effectiveness.

Current research in AR-based education mainly emphasizes **static content visualization, pre-recorded instructional explanations, and limited user interaction**. Most systems deliver predefined learning materials without adapting to individual learner behavior or understanding levels. As a result, students receive the same learning experience regardless of their performance, pace, or learning style.

Several critical gaps have been identified in existing solutions. Most platforms do not support **gesture-based contextual interaction**, preventing natural and intuitive user engagement. Additionally, there is an absence of **voice-based intelligent assistants** that allow learners to ask questions in real time. Existing systems also lack **AI-generated quizzes** and **adaptive learning analytics**, which are essential for personalized assessment and feedback. Furthermore, **gamification elements** such as rewards, progress tracking, and adaptive challenges are either minimal or completely absent.

Due to these limitations, current AR-based learning systems fail to analyze student behavior or provide customized feedback. They also do not integrate real-time AR content with conversational artificial intelligence, which restricts their ability to offer interactive and context-aware learning experiences. This highlights the need for a more intelligent and adaptive AR-based STEM learning platform.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

This project proposes an **AR-Based Intelligent STEM Learning System** designed to enhance student engagement and conceptual understanding through immersive and adaptive learning technologies. The system integrates four core functional modules that collectively address the limitations identified in existing educational platforms.

The first module focuses on **AR-Based STEM Visualization**, where static textbook content is transformed into interactive three-dimensional models. The system accepts inputs such as a book page QR code and the learner's hand gestures. Using AR processing techniques, relevant 3D STEM models are loaded and displayed in real time. Students can rotate, zoom, and isolate specific components through intuitive gesture interactions, allowing them to explore complex structures such as human organs, planetary systems, and physics experiments in an immersive environment. This improves spatial understanding and knowledge retention.

The second module is a **Context-Aware Voice-Based Learning Assistant** that enables natural communication between students and the system. Students can ask questions verbally while interacting with AR content. The system processes speech input through speech-to-text conversion and applies AI-based intent analysis to generate relevant responses. The output is provided as spoken explanations accompanied by visual highlights within the AR environment. This feature ensures that responses are contextually aligned with the content currently viewed by the learner, creating a personalized and interactive learning experience.

The third module includes an **AI-Generated Quiz and Smart Learning Analytics** component. After each lesson, the system automatically generates quizzes based on the content covered. Student responses and response times are analyzed to evaluate understanding, confidence levels, and learning patterns. Based on this analysis, personalized feedback is provided to support continuous improvement. This adaptive assessment mechanism allows teachers and parents to monitor student progress more effectively.

The final module introduces **Gamified STEM Missions** to motivate learners and encourage continuous engagement. Students earn experience points, badges, and rewards for completing tasks and achieving learning milestones. The system dynamically adjusts mission difficulty based on student performance, ensuring an optimal challenge level. This gamification approach fosters motivation, promotes skill mastery, and enhances long-term learning commitment.

Overall, the proposed system integrates augmented reality, artificial intelligence, and gamification to create a comprehensive and intelligent STEM learning platform. By combining immersive visualization, conversational assistance, adaptive assessment, and game-based motivation, the solution effectively addresses existing gaps and supports

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

To acquire the necessary domain expertise, team members will engage in continuous learning by following **Augmented Reality (AR) and Unity development tutorials**, studying **artificial intelligence and large language model (LLM) integration**, and learning to use **speech processing APIs** for voice-based interaction. In addition, members will participate in relevant **workshops, webinars, and online training programs** to enhance both technical and domain-specific knowledge.

The technical skill set required for the project includes **Unity for AR development, AR Foundation for cross-platform AR support, MediaPipe for gesture recognition, OpenAI APIs for intelligent content generation, and React and Node.js for system integration and interface development**. Team members will practice through hands-on projects and guided labs to strengthen these skills.

Data required for the system will be collected through **student quiz responses, interaction logs, and voice-based questions**. All data collection will be conducted ethically and solely for research and system improvement purposes.

Since the project involves **children's data**, ethical clearance is mandatory. Informed consent will be obtained from **parents, schools, and relevant authorities** prior to data collection. All personal data will be **anonymized**, securely stored, and accessed only by authorized members. Data handling will strictly follow ethical research guidelines to ensure **privacy, confidentiality, and security**.

9. Objectives and Novelty

Main Objective
To develop an intelligent AR-based STEM learning platform that enhances engagement and understanding among primary students.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22087638 ELWITIGALA S A I	AR-Based STEM Visualization	• Develop 3D STEM models • Implement QR-based content detection • Integrate hand gesture controls (rotate, zoom, isolate) • Optimize AR performance	*Existing Problems* - Current AR educational apps use fixed, universal gestures across all content, causing cognitive overload. Students use the same gesture for rotating a cell and orbiting a planet, creating conceptual confusion and limiting natural interaction with complex 3D models. *Novel Approach:-* Dynamic gesture mapping that changes meaning based on lesson context. The same pinch gesture isolates cell components in biology but adjusts orbit speed in astronomy. QR-based lesson detection automatically loads subject-specific gesture schemas. *Expected Data Output-* Gesture recognition accuracy per lesson type, student interaction heatmaps showing context-specific gesture preferences, context-switching success rates when transitioning between subjects, and time-to-task completion improvements across STEM domains.

IT22136756 MANODYA K M B	Context-Aware Voice Assistant	• Integrate speech-to-text APIs • Develop intent recognition logic • Connect AI responses with AR content • Generate spoken answers • Highlight objects in AR during explanations	*Existing Problems* - Traditional voice assistants operate independently of visual content, providing generic answers without knowing what students are viewing. This forces mental bridging between audio explanations and AR visuals, reducing learning efficiency. *Novel Approach*- time AR view awareness that references and highlights specific objects students are viewing. Responds with "The mitochondria *you're looking at* produces energy" while automatically isolating and animating that component during explanation. *Expected Data Output*- Query-to-AR-context matching accuracy measuring spatial reference resolution, student clarification request frequency indicating reduced confusion, voice-to-AR synchronization latency, and gaze-pattern correlation showing whether synchronized explanations improve attention and comprehension.

IT22272768 BASNAYAKA W B M D K	AI Quiz & Learning Analytics	• Design prompt-based quiz generation system • Implement dynamic question generation based on lesson content • Analyze response accuracy and time • Generate personalized feedback reports • Create performance dashboards	*Existing Problems* - Conventional assessments focus only on answer correctness, missing hesitation patterns and confidence levels. Students who guess correctly get the same score as those with deep understanding, failing to identify who needs intervention despite passing grades. *Novel Approach*- Analyzes behavioral metadata (response time distributions, answer revisions, confidence indicators) alongside correctness. Machine learning distinguishes guessing from mastery, generating feedback on hesitant correct answers and identifying struggling students hidden by lucky guesses. *Expected Data Output*- Confidence score estimates from behavioral cues, knowledge gap identification maps showing partial versus complete mastery, learning curve progression tracking skill development patterns, and personalized intervention recommendations based on misconception clustering rather than just wrong answers.

IT22292568 DILSHAN W A	Gamified Missions	• Design integrated skill mastery tracking system • Implement hands-on AR mission creation • Develop adaptive mission unlocking logic • Build comprehensive progress and achievement dashboard	*Existing Problems* - Traditional gamification rewards linear content completion rather than actual learning. Students complete lessons sequentially without demonstrating practical application of learned concepts. Gamified elements exist as separate entertainment layers disconnected from the actual learning journey *Novel Approach*-- Integrates the entire learning pipeline—AR exploration, voice inquiry, quiz performance, and hands-on AR missions—into unified skill mastery tracking. Students progress by applying what they learned through interactive AR challenges (e.g., assembling a cell after studying it, building solar systems after exploring planets). Unlocks are based on demonstrated practical application across all prior components, not individual task completion. *Expected Data Output*- Integrated learning journey completion rates tracking progression from exploration to practical application, skill retention metrics comparing knowledge before quizzes versus after hands-on missions, cross-component performance correlation showing how AR interaction and voice engagement predict practical task success, and holistic mastery scores aggregating performance across visualization, inquiry, assessment, and application phases.

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

Need more improvements at the proposal level.

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Dharshana	Ginige	
Co-Supervisor	Mr	Jeewaka	Perera	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				